How to work

- Use the same course folder as in week 1. Add the two new data files to `data/` and the two new worksheets to `do/exercises/`. In Stata, move to the folder with `cd "path/to/folder"`.
- Open the worksheet, save it under a new name, and complete the TODO lines. Write your interpretations as comments.
- **Predict before you run**: for every PREDICT line, write your guess before executing the command.
- Work in pairs, swapping who types and who explains. Every number comes with its unit and the number of observations behind it.
- Laboratory work is formative and not graded.

Data for week 2

- `data/justice_europe_2019.dta`: 37 European countries, one row per country, 2019. Professional judges and police officers per 100,000 inhabitants; police-recorded offences per 100,000 (homicide, robbery, burglary, theft, fraud, corruption). Source: Eurostat, datasets `crim_just_job` and `crim_off_cat` (free reuse, source acknowledged; selected and merged by the instructor). Merged with Doing Business 2020 enforcement time and judicial-quality index, and with World Bank GDP per capita 2019.
- `data/gdp_2019.dta`: 210 economies. GDP (current US$) and population 2019, World Bank World Development Indicators (CC BY 4.0); GDP per capita computed as their ratio.
- `data/contracts_2019.dta`: the Doing Business file of week 1.

Recorded offences count what is reported to and recorded by the police, under national legal definitions; they are not measures of how much crime occurs.

Lab 3: relationships and tests

Questions. Do more judges make courts faster? Does more police mean less crime? Are richer economies faster at enforcing contracts?

Worksheet. `do/exercises/03_relationships_and_tests.do`.

Tasks

1. What is one row of `justice_europe_2019`? Which variables are missing, and for which countries?
2. Scatter and correlate enforcement time with judges per 100,000 (`pwcorr ..., sig obs`). Give one reverse-causality story and one confounder.
3. Scatter and correlate recorded theft with police per 100,000, then add log GDP per capita. What could explain the correlation other than deterrence?
4. Rank countries by recorded corruption. Why need a high value not mean more corruption?
5. In the 191 Doing Business economies, test whether high-income economies are faster (`ttest`, with and without `unequal`). Report difference, p-value and 95% interval in one correct sentence.

6. Compare the share of economies with any court automation by income group (`prtest`). Does this show that automation makes courts faster?
7. Write three sentences: one association; one alternative explanation; one test result.

Extension. Correlate police with six crime categories. How many correlations have $p < 0.05$, and are they independent pieces of evidence?

Self-check

- Did I report n for every correlation (pairs with both values non-missing)?
- Did I avoid writing that the p-value is the probability that the null hypothesis is true?
- Did I avoid reading "not significant" as "no difference"?

Key definitions

- Correlation: $r_{xy} = \frac{1}{n-1} \sum_i \frac{(x_i - \bar{x})}{s_x} \frac{(y_i - \bar{y})}{s_y}$, between -1 and 1, linear association only, sensitive to extreme values.
- Test of a difference in means: $t = (\bar{y}_0 - \bar{y}_1)/\mathrm{SE}(\bar{y}_0 - \bar{y}_1)$. The p-value is the probability, if the null hypothesis were true, of a t at least as far from zero as the one observed.
- Three alternative explanations of an association: reverse causality, confounding, measurement. Chance is a fourth; tests address only this one.
- Internal validity: is the comparison credible in the units studied? External validity: would the result hold elsewhere?

Lab 4: simple regression

Question. Italy takes 1,120 days to enforce a contract. Is that slow for an economy as rich as Italy?

Worksheet. `do/exercises/04_simple_regression.do`.

Before class: reading

Voigt, S. and Wulf, A. J. (2019), What makes prosecutors independent? Analysing the institutional determinants of prosecutorial independence, *Journal of Institutional Economics* 15(1): 99–120, doi:10.1017/S1744137417000212. Read the abstract, Sections 1 and 5, Section 7 and the first two paragraphs of Section 8 (pp. 112–113), Table 1 and column (1) of Table 2. Bring notes on:

1. What is one observation in their data?
2. How are *de jure* (law on paper) and *de facto* (practice) independence measured? Why might they diverge?
3. One coefficient in column (1) of Table 2: its sign, the unit of the explanatory variable, its standard error, and one sentence in words. For a 0/1 variable such as English legal origin: compared with which countries?
4. Footnote 10 reads a non-significant normality test as showing that the data are normal. Is that a correct reading of a p-value?

Tasks

1. Merge `gdp_2019` into `contracts_2019` (`merge 1:1 economy_code`). How many economies are matched? Which are lost, and are they a random subset?
2. Regress log time on log GDP per capita. Write the slope as a sentence with units; what does doubling income imply?
3. Compute predictions and residuals. Report Italy's predicted and observed days and its residual. Name two things the residual contains other than "inefficiency".
4. Regress time on the quality of judicial processes index. Slope in days per point, 95% interval, R^2.
5. Regress time on a high-income 0/1 variable and compare with `ttest`. Which numbers do the constant and the slope reproduce?
6. In Europe, regress time on judges per 100,000, with and without Liechtenstein. Would you advise a ministry on this basis?
7. Paper bridge: one coefficient from Voigt and Wulf in words. Is it causal?

Extension. Regress recorded theft on police, then add log GDP per capita. What happens to the police coefficient?

Self-check

- Did I check `_merge` before dropping anything?
- Did I write every slope with the units of both variables?
- Did I describe the residual as a question to investigate, not as a verdict?

Reading slopes with logarithms

Model	A change in x of	goes with a change in y of about
y on x	+1 unit	$\hat{\beta}_1$ units
y on $\ln x$	+1%	$\hat{\beta}_1/100$ units
$\ln y$ on x	+1 unit	$100\,\hat{\beta}_1\%$
$\ln y$ on $\ln x$	+1%	$\hat{\beta}_1\%$ (elasticity)

For large changes compute exactly: in a log-log model, multiplying x by k multiplies the prediction of y by $k^{\hat{\beta}_1}$. A regression on a 0/1 variable reproduces two group means: the constant is the mean of group 0, the slope is the difference between the means.